Digital signatures

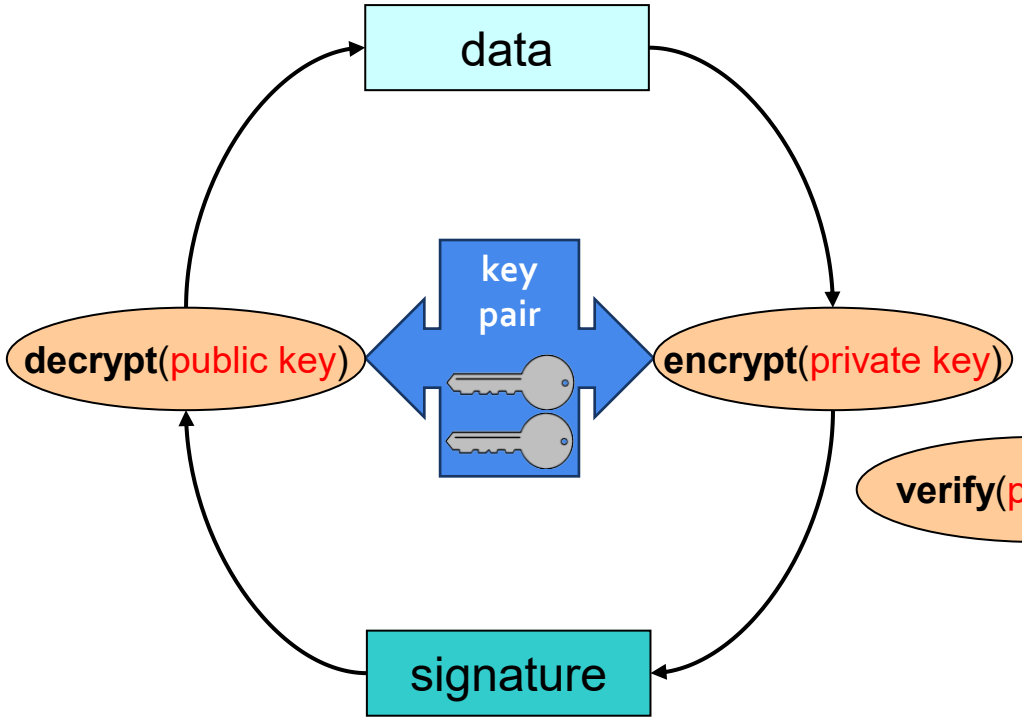
Asymmetric (Block) Ciphers

- Use key pairs
  - One private key (personal, not transmittable)
  - One public key, available to all

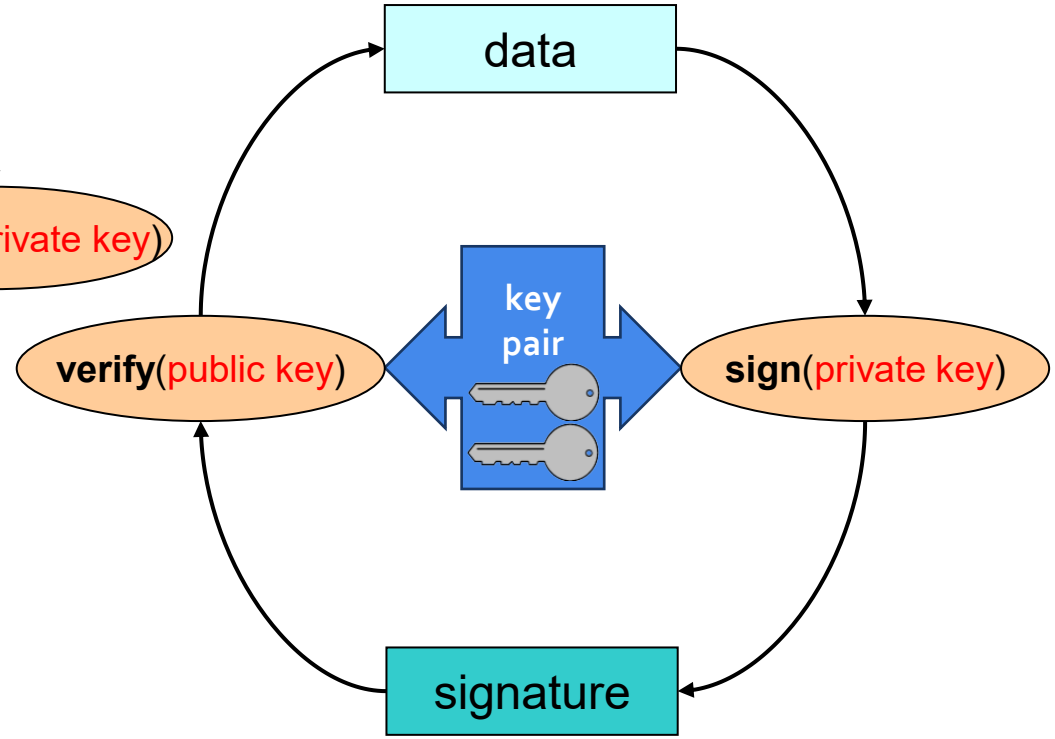
- Allow
  - Confidentiality without any previous exchange of secrets
  - Authentication
    - Of contents (data integrity)
    - Of origin (source authentication, or digital signature)

Digital signatures

Encrypt / decrypt (RSA)



Sign / verify (ElGamal, EC)



Digital signatures

- Authenticate the contents of a document
  - Ensure its integrity (it was not changed)
- Authenticate its author
  - Ensure the identity of the creator/originator
- Prevent the repudiation of signatures
  - Non-repudiation
  - Genuine authors cannot deny authorship
    - Only the identified author could have generated a given signature

Digital Signatures: approaches

- Asymmetric encryption / decryption
- Asymmetric signature / verification
- Digest functions
  - Only for performance

Signing:

$A_x(doc) = info + Enc(K_x^{-1}, digest(doc + info))$

$A_x(doc) = info + Sign(K_x^{-1}, digest(doc + info))$

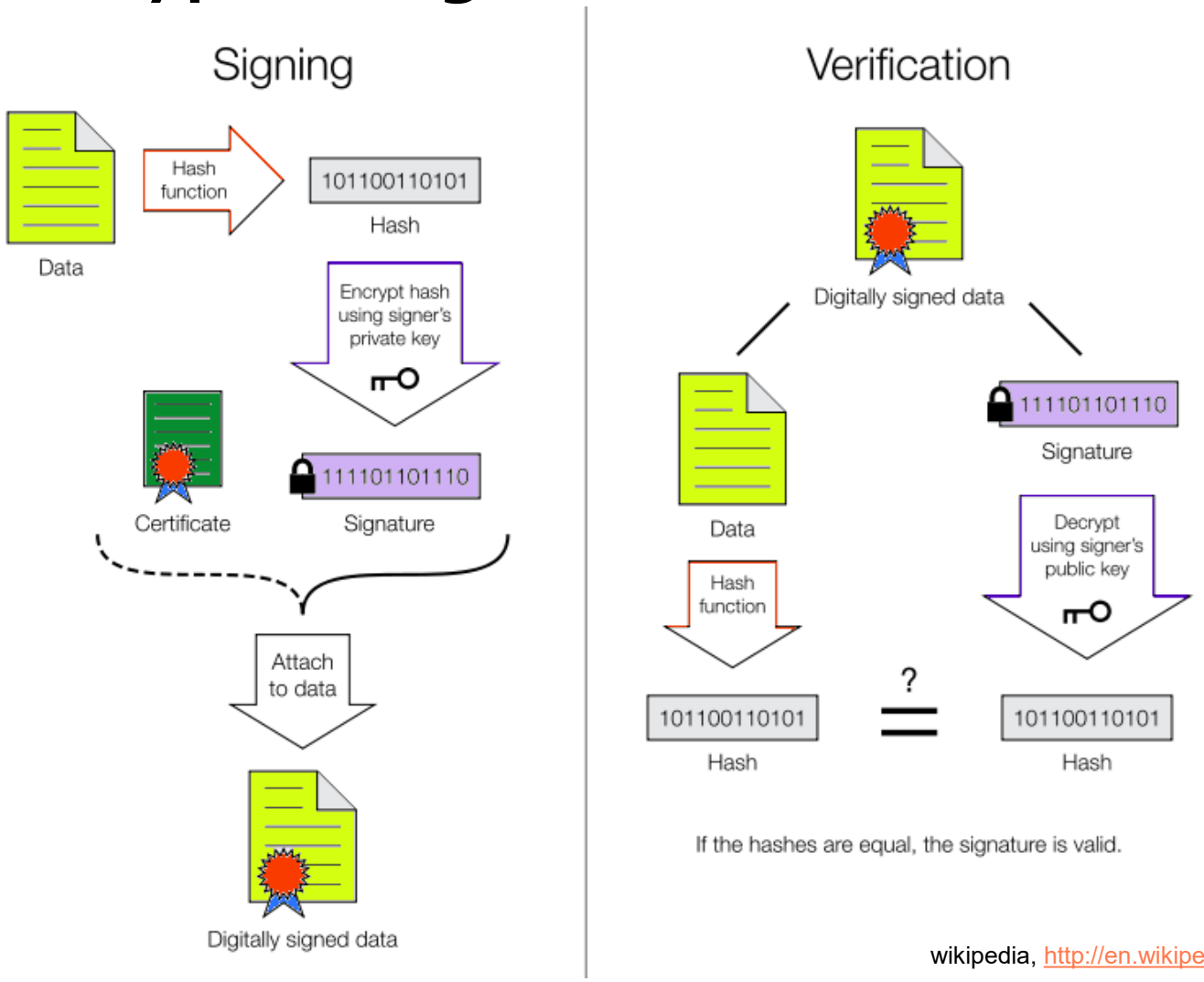
$info$ = signing context, signer identity, K_x

Verification:

$\mathbf{D(K_x, A_x(doc)) \equiv digest(doc + info)}$

$\mathbf{V(K_x, A_x(doc), doc, info)} \rightarrow$ True / False

Encryption / decryption signatures



wikipedia, http://en.wikipedia.org/wiki/Digital_signature

Digital signature on a mail: Multipart content, signature w/ certificate

```
From - Fri Oct 02 15:37:14 2009
[...]
Date: Fri, 02 Oct 2009 15:35:55 +0100
From: =?ISO-8859-1?Q?Andr=E9_Z=FAquete?= <andre.zuquete@ua.pt>
Reply-To: andre.zuquete@ua.pt
Organization: IEETA / UA
MIME-Version: 1.0
To: =?ISO-8859-1?Q?Andr=E9_Z=FAquete?= <andre.zuquete@ua.pt>
Subject: Teste
Content-Type: multipart/signed; protocol="application/x-pkcs7-signature"; micalg=sha1; boundary="------------ms050405070101010502050101"

This is a cryptographically signed message in MIME format.

--------------ms050405070101010502050101
Content-Type: multipart/mixed;
 boundary="------------060802050708070409030504"

This is a multi-part message in MIME format.
--------------060802050708070409030504
Content-Type: text/plain; charset=ISO-8859-1
Content-Transfer-Encoding: quoted-printable

Corpo do mail

--------------060802050708070409030504—
--------------ms050405070101010502050101
Content-Type: application/x-pkcs7-signature; name="smime.p7s"
Content-Transfer-Encoding: base64
Content-Disposition: attachment; filename="smime.p7s"
Content-Description: S/MIME Cryptographic Signature

MIAGCSqGSIb3DQEHAqCAMIACAQExCzAJBgUrDgMCGgUAMIAGCSqGSIb3DQEHAQAAoIIamTCC
BUkwggSyoAMCAQICBAcnIaEwDQYJKoZIhvcNAQEFBQAwdTELMAkGA1UEBhMCVVMxGDAWBgNV
[...]
KoZIhvcNAQEBBQAEgYCofks852BV77NVuww53vSxO1XtI2JhC1CDlu+tcTPoMD1wq5dc5v40
TgsawoN8dqgVLk8aC/CdGMbRBu+J1LKrcVZa+khnjjtB66HhDRLrjmEGDNttrEjbqvpd2QO2
vxB3iPTlU+vCGXo47e6GyRydqTpbqor49Zqmx+IJ6Z7iigAAAAAAAA==
--------------ms050405070101010502050101--
```

Bibliografia

Leitura recomendada:

- Segurança em Redes Informáticas, A. Zúquete: Chap. 2

Leitura adicional:

- Digital signature
- Data integrity
- Message authentication
- Non-repudiation
- Elliptic Curve Digital Signature Algorithm (ECDSA)